



IT 6/L (8419) – Fundamentals of Database System

NullCRM: Customer Relationship Management System

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I. Project Context

The NullCRM is a customer management system designed to help businesses and organizations centralize and manage customer information in a simple and efficient way. The system allows users to store, update, and view vital records such as contact details, sales leads, communication histories, and follow-up schedules. It aims to replace fragmented manual processes and disorganized spreadsheets that often lead to missed opportunities and slow response times.

This project focuses on improving how sales and customer service tasks are handled. By using this system, staff can easily track the progress of potential leads, log every customer interaction, and manage pending tasks without confusion. It also ensures that all data is kept in one secure location, making it easier to maintain accuracy and update information in real time.

The system is created with a user-friendly interface so that users can navigate and manage data without difficulty. It is designed for businesses that need a reliable, role-based platform to organize their sales pipeline and monitor team performance. Overall, NullCRM helps save time, improve lead conversion rates, and support better decision-making through clear, data-driven reporting.

II. Problem Statement

The current methods of managing customer relations within the organization are hindered by significant operational challenges, primarily centered around data fragmentation and inefficiency. Without a centralized

system, customer information and communication logs are often scattered across disparate tools, making it nearly impossible to maintain a consistent "single source of truth" or track the progression of leads effectively. This lack of integration leads to poor task accountability, as scheduling follow-ups and monitoring staff assignments becomes a manual, error-prone process. Furthermore, the absence of a structured, role-based authentication system creates security risks and makes it difficult to enforce proper data access levels between sales staff and management.

III. Proposed Solution

To address these issues, the development of NullCRM, a Laravel-based web application, is proposed to centralize and automate the organization's sales and customer management workflows. By leveraging a relational database and the Eloquent ORM, the system provides a unified platform for storing all customer profiles, lead statuses, and detailed interaction logs in one secure location. NullCRM introduces dedicated modules for lead tracking and follow-up management, ensuring that every sales opportunity is monitored from inception to completion through a structured digital pipeline. Additionally, the implementation of role-based access control (RBAC) ensures that sensitive data is protected, while the reporting module offers managers real-time insights into team performance and sales activities.

IV. Objectives of the Study

This study aims to develop a CRM system with the following objectives:

- Manage customer profiles and contact information
- Record and monitor sales leads and opportunities
- Track follow-ups, meetings, and customer interactions
- Assign roles and permissions for system users
- Generate summary reports for customer and sales activities

V. Scope and Delimitation

The scope of this project centers on the development of a comprehensive Customer Relationship Management (CRM) system that facilitates secure user authentication and authorization. The application is designed to streamline customer management and lead management, providing a centralized platform for tracking potential clients and existing accounts. It incorporates robust activity and interaction logging to maintain a detailed history of communications, alongside a follow-up scheduling module to manage upcoming tasks and reminders. Furthermore, the system includes a dashboard and reporting feature for real-time performance monitoring, all protected by role-based access control to ensure data integrity and security across different user levels.

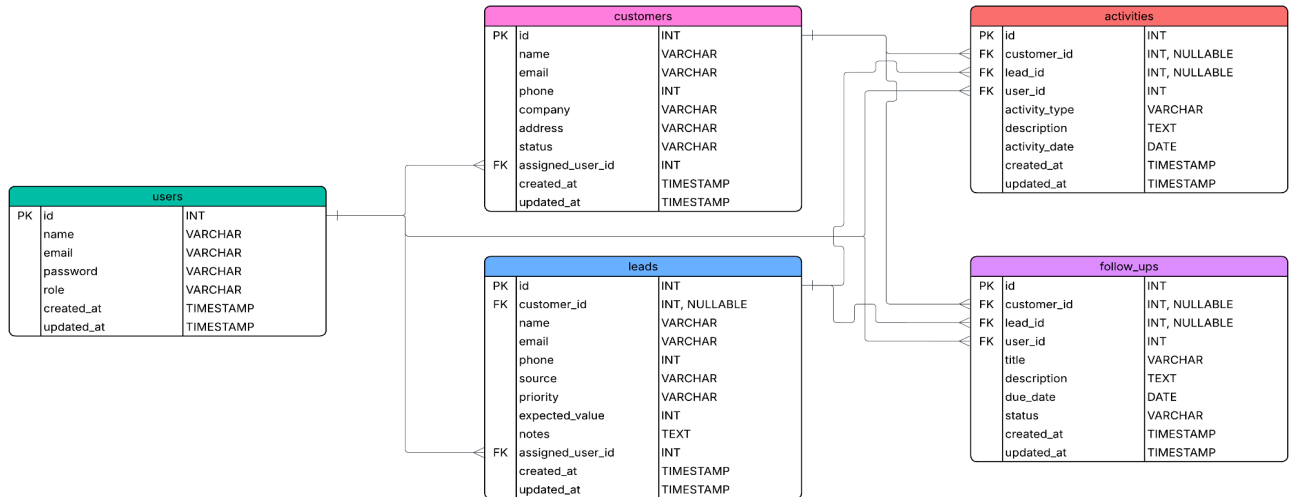
VI. System Analysis and Design

a. Business Rules

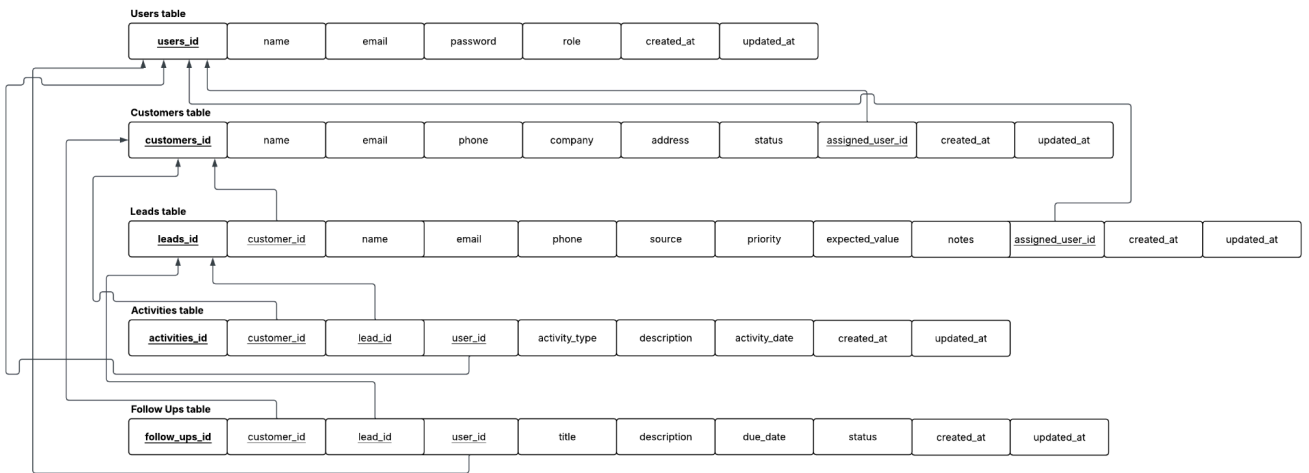
The system will be implemented with the following business rules:

- Only authenticated users can access the system
- Only admins can manage user accounts
- A customer may be assigned to only one staff member at a time
- A lead must have a valid status
- A follow-up must have a due date
- Completed follow-ups cannot be edited unless reopened
- Every activity must be linked to a user
- Deleting a customer should also handle related records properly
- A lead may be converted into a customer
- Reports must only show data allowed by user role

b. ERD



c. Mapping Entities to Relations



d. Entity Relationships

- One user can manage many customers
- One user can manage many leads
- One customer can have many activities

- One lead can have many activities
- One customer can have many follow-ups
- One lead can have many follow-ups
- One user can create many activities and follow-ups

VII. Database Design

a. List of Tables

Server: 127.0.0.1 » Database: laravel

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ activities	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
☐ cache	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ cache_locks	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ customers	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ follow_ups	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
☐ leads	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ migrations	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ password_reset_tokens	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ sessions	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
10 tables	Sum	22	InnoDB	utf8mb4_general_ci	400.0 KiB	0 B

☐ Check all

[Print](#)
[Data dictionary](#)

[Create new table](#)

Console

Note: The tables that are highlighted in the list such as the cache, sessions, jobs, migrations, and the password_reset_tokens are Laravel framework system tables and are not part of the core application design.

b. Fields, Data Types, Constraints

Users Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 email	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 email_verified_at	timestamp			Yes	NULL			Change Drop More
☐	5 password	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	6 remember_token	varchar(100)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 role	varchar(255)	utf8mb4_unicode_ci		No	sales_staff			Change Drop More
☐	8 created_at	timestamp			Yes	NULL			Change Drop More
☐	9 updated_at	timestamp			Yes	NULL			Change Drop More

Customers Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 first_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 last_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 email	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	5 phone	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	6 company_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	7 address	text	utf8mb4_unicode_ci		No	None			Change Drop More
☐	8 status	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	9 assigned_user_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	10 created_at	timestamp			Yes	NULL			Change Drop More
☐	11 updated_at	timestamp			Yes	NULL			Change Drop More
☐	12 deleted_at	timestamp			Yes	NULL			Change Drop More

Leads Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 customer_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	3 name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 email	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	5 phone	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	6 source	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	7 status	enum('New', 'Contacted', 'Qualified', 'Proposal Sent')	utf8mb4_unicode_ci		No	New			Change Drop More
☐	8 priority	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	9 expected_value	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	10 notes	text	utf8mb4_unicode_ci		No	None			Change Drop More

Activity Table

Follow Ups Table

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id 	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	 Change  Drop  More
☐	2	customer_id 	bigint(20)		UNSIGNED	Yes	NULL			 Change  Drop  More
☐	3	lead_id 	bigint(20)		UNSIGNED	Yes	NULL			 Change  Drop  More
☐	4	user_id 	bigint(20)		UNSIGNED	No	None			 Change  Drop  More
☐	5	title	varchar(255)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	6	description	text	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	7	due_date	date			No	None			 Change  Drop  More
☐	8	status	varchar(255)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	9	created_at	timestamp			Yes	NULL			 Change  Drop  More
☐	10	updated_at	timestamp			Yes	NULL			 Change  Drop  More
☐	11	deleted_at	timestamp			Yes	NULL			 Change  Drop  More

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id 	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	 Change  Drop  More
☐	2	customer_id 	bigint(20)		UNSIGNED	Yes	NULL			 Change  Drop  More
☐	3	lead_id 	bigint(20)		UNSIGNED	Yes	NULL			 Change  Drop  More
☐	4	user_id 	bigint(20)		UNSIGNED	No	None			 Change  Drop  More
☐	5	activity_type	varchar(255)	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	6	description	text	utf8mb4_unicode_ci		No	None			 Change  Drop  More
☐	7	activity_date	date			No	None			 Change  Drop  More
☐	8	created_at	timestamp			Yes	NULL			 Change  Drop  More
☐	9	updated_at	timestamp			Yes	NULL			 Change  Drop  More
☐	10	deleted_at	timestamp			Yes	NULL			 Change  Drop  More

VIII. Advanced Database Features

a. Triggers

The following triggers are implemented in the database to automate critical operations and maintain data consistency.

Trigger 1: after_lead_status_update

The purpose of this is to maintain an automated audit trail of all lead progress by logging every status change as a new entry in the activity feed.

Explanation: This trigger fires after an update on the **leads** table specifically when the status column is modified. It automatically

generates a new record in the **activities** table, capturing the lead ID and the assigned user. The activity is categorized as a "Status Update," and the description is dynamically generated to show exactly what the status was changed from and what it is now (e.g., "Status changed from New to Qualified"). It also logs the current date and timestamps for the audit record.

Details

Trigger name

after_lead_status_update

Table

leads

Time

AFTER

Event

UPDATE

Definition

```
1 BEGIN
2   IF OLD.status <> NEW.status THEN
3     INSERT INTO activities (lead_id, user_id,
4       activity_type, description, activity_date, created_at,
5       updated_at)
6     VALUES (NEW.id, NEW.assigned_user_id, 'Status
7       Update',
8         CONCAT('Status changed from ', OLD.status,
9           ' to ', NEW.status),
10        CURDATE(), NOW(), NOW());
11   END IF;
12 END
```

Go

Close

Trigger 2: sync_customer_on_won_lead

This is to automatically synchronize the status of a linked customer profile to "Active" once a lead has been successfully converted to a "Won" status.

Explanation: This trigger fires after an update on the **leads** table. It checks if the lead's status has been changed to 'Won' and ensures there is a valid `customer_id` associated with the record. Once these conditions are met, it automatically updates the corresponding row in the **customers** table, setting their status to 'Active' and refreshing the `updated_at` timestamp to reflect the recent conversion.

Details

Trigger name

sync_customer_on_won_le

Table

leads

Time

AFTER

Event

UPDATE

Definition

```
1 BEGIN
2   IF NEW.status = 'Won' AND NEW.customer_id IS NOT NULL
3   THEN
4     UPDATE customers SET status = 'Active',
5       updated_at = NOW()
6     WHERE id = NEW.customer_id;
7   END IF;
8 END
```

Go

Close

Trigger 3: before_follow_up_insert

This is to prevent the creation of follow-up tasks with deadlines that have already passed, ensuring data integrity within the scheduling system.

Explanation: This trigger fires before a new record is inserted into the follow_ups table. It performs a validation check comparing the due_date of the new task against the current system date (CURDATE()). If the provided date is in the past, the trigger uses a SIGNAL SQLSTATE to abort the insertion and returns a custom error message: "Due date cannot be in the past."

Details

Trigger name

before_follow_up_insert

Table

follow_ups

Time

BEFORE

Event

INSERT

b. Views

View 1: staff_workload

Combines users, customers, and leads to see how much each staff member is handling.

Structure:

```
1 CREATE OR REPLACE VIEW staff_workload AS
2 SELECT u.name AS staff_name,
3        COUNT(DISTINCT c.id) AS active_customers,
4        COUNT(DISTINCT l.id) AS total_leads
5 FROM users u
6 LEFT JOIN customers c ON u.id = c.assigned_user_id
7 LEFT JOIN leads l ON u.id = l.assigned_user_id
8 GROUP BY u.id;
```

View 2: urgent_tasks

Shows follow_ups that are 'pending' but past their due_date, linked to the users table.

Structure:

```
CREATE OR REPLACE VIEW urgent_tasks AS
SELECT f.title, f.due_date, u.name AS assigned_to, l.name AS lead_name
FROM follow_ups f
JOIN users u ON f.user_id = u.id
LEFT JOIN leads l ON f.lead_id = l.id
WHERE f.status = 'pending' AND f.due_date < CURDATE()
AND f.deleted_at IS NULL;
```

View 2: pipeline_financials

Summarizes the potential money in the leads table based on the current pipeline status.

Structure:

```
CREATE OR REPLACE VIEW pipeline_financials AS
SELECT status,
       COUNT(*) AS lead_count,
       SUM(CAST(expected_value AS DECIMAL(10,2))) AS potential_revenue
FROM leads
WHERE deleted_at IS NULL
GROUP BY status;
```

IX. System Features and Functionality

The system will have the following modules:

a. Authentication Module

Description: Handles login, logout, profile access, and authorization.

Features:

- User login
- User logout
- Password hashing
- Role-based middleware
- Profile management

b. Dashboard Module

Description: Displays system summaries and activity snapshots.

Features:

- Total customers
- Total active leads
- Total completed follow-ups
- Recent activities
- Upcoming follow-ups

c. Customer Module

Description: Stores and manages customer records.

Features:

- Add customer
- Edit customer
- Delete customer

- View customer details
- Search and filter customers
- Assign customer to staff

d. Lead Module

Description: Tracks potential clients and sales opportunities.

Features:

- Add lead
- Update lead status
- Convert lead to customer
- Assign lead to staff
- Set priority
- Record expected deal value

e. Activity Module

Description: Records communications and actions related to a customer or lead.

Features:

- Log calls
- Log emails
- Log meetings
- Add notes
- View activity history per customer or lead

f. FollowUp Module

Description: Schedules reminders and next actions for customers and leads.

Features:

- Create follow-up task
- Set due date
- Mark as completed
- View pending follow-ups
- Assign follow-up to user

g. Report Module

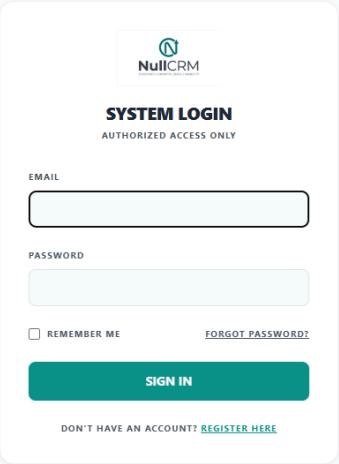
Description: Generates summary reports for monitoring performance and activity.

Features:

- Total customers report
- Lead status report
- Sales pipeline summary
- User activity report
- Follow-up completion report

X. Prototype / User Interface

a. Login Module



The image shows a 'SYSTEM LOGIN' form for 'NullCRM'. The form is centered on a light blue background. It includes fields for 'EMAIL' and 'PASSWORD', a 'REMEMBER ME' checkbox, a 'FORGOT PASSWORD?' link, a 'SIGN IN' button, and a 'REGISTER HERE' link at the bottom.

NullCRM

SYSTEM LOGIN
AUTHORIZED ACCESS ONLY

EMAIL

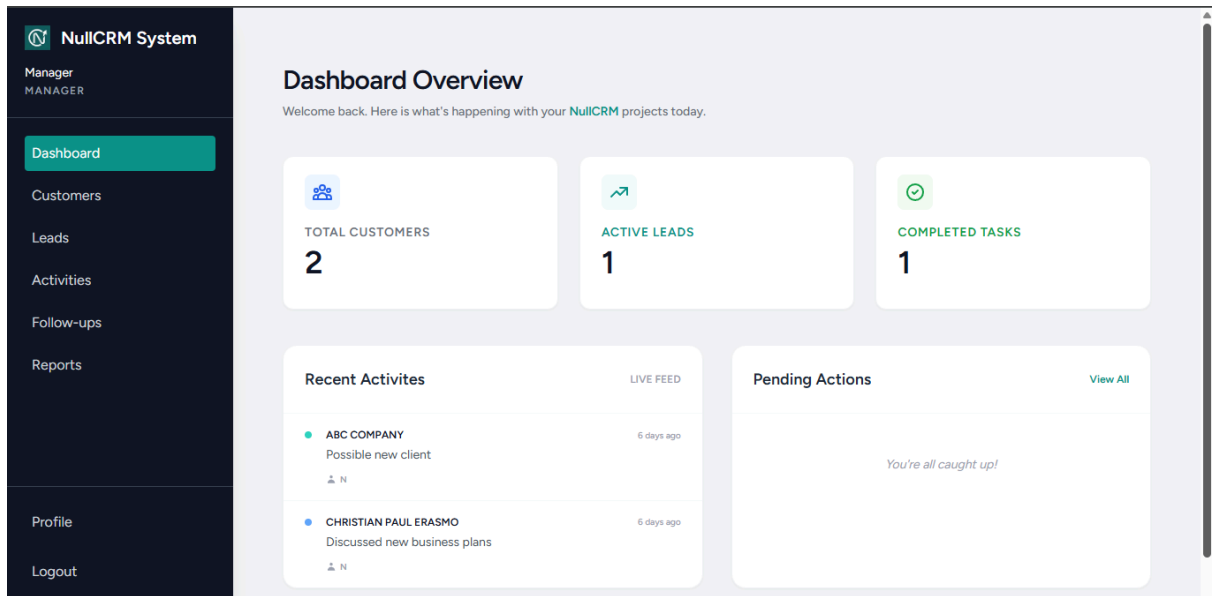
PASSWORD

☐ REMEMBER ME [FORGOT PASSWORD?](#)

SIGN IN

DON'T HAVE AN ACCOUNT? [REGISTER HERE](#)

b. Dashboard Module



c. Customer Module

Customer Index

The screenshot shows the 'Customer Index' section of the NullCRM System. The left sidebar contains navigation links: Dashboard, Customers (selected), Leads, Activities, Follow-ups, Reports, Profile, and Logout. The main content area features a 'Customer Index' header with a '+ Add Customer' button, a search bar, and status filters for 'Active: 2' and 'Inactive: 0'. Below this are two sections: 'ACTIVE CONTACTS' and 'INACTIVE CONTACTS'. The 'ACTIVE CONTACTS' section contains a table with two rows of customer data. The 'INACTIVE CONTACTS' section contains a message 'No inactive contacts found.'.

NullCRM System
Manager
MANAGER

Customer Index

[+ Add Customer](#) Active: 2 Inactive: 0

ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmo	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	View Edit Delete
Ulysses Kaiser Alzate	uka@gmail.com	09098152500	CPE COMPANY	Sales SALES_STAFF	View Edit Delete

INACTIVE CONTACTS

No inactive contacts found.

Add Customer

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

← Add New Customer

CONTACT INFORMATION

FIRST NAME
e.g. John

LAST NAME
e.g. Doe

EMAIL ADDRESS
john.doe@example.com

CONTACT NUMBER
09123456789

COMPANY & CLASSIFICATION

COMPANY NAME
e.g. ABC Corp

STATUS
Active

OFFICE ADDRESS
Street, City, Province

ASSIGN TO
-- Select Assignee --

Edit Customer

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

← Christian Paul Erasmo
CUSTOMER PROFILE • ID #1

ACTIVE

GENERAL INFORMATION

EMAIL ADDRESS
cpe@gmail.com

CONTACT NUMBER
09694201500

COMPANY / ORGANIZATION
CPE Company

OFFICE ADDRESS
Narnia

INTERNAL ASSIGNMENT DETAILS

ASSIGNED STAFF NAME
Manager

STAFF USER ID
#3

RECENT ACTIVITIES

+ LOG NEW ACTIVITY

MEETING 2026-04-30

Activity Logged
Discussed new business plans

STAFF: N

Delete Customer

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

+ Add Customer

127.0.0.1:8000 says
Delete this contact?

OKCancel

Active: 2Inactive: 0

▼ ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmo	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	  
Ulysses Kaiser Alzate	uka@gmail.com	09098152500	CPE COMPANY	Sales SALES_STAFF	  

▼ INACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
No inactive contacts found.					

d. Lead Module

Lead Index

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

+ Add New Lead

FILTER BY STATUS: All Leads

Search name or source...

PROSPECT	STATUS	PRIORITY	VALUE	ASSIGNED TO	ACTIONS
ABC Company STOCKS	NEW	MEDIUM	₱50,000.00	Sales SALES_STAFF	   

Add Lead

NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

← New Sales Opportunity

OPPORTUNITY DETAILS

CUSTOMER OR PROSPECT NAME

EMAIL ADDRESS

example@mail.com

PHONE NUMBER

09123456789

SOURCE

e.g. Website, Referral

STATUS

New

PRIORITY

Medium

EXPECTED VALUE (P)

ASSIGN TO

-- Select Assignee --

NOTES

Edit Lead

NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

← ABC Company

LEAD OPPORTUNITY • SOURCE: STOCKS

NEW

OPPORTUNITY DETAILS

EXPECTED DEAL VALUE

₱50,000.00

PRIORITY LEVEL

MEDIUM

EMAIL ADDRESS

abc@gmail.com

CONTACT NUMBER

09123456789

INTERNAL NOTES & CONTEXT

N/A

INTERNAL ASSIGNMENT

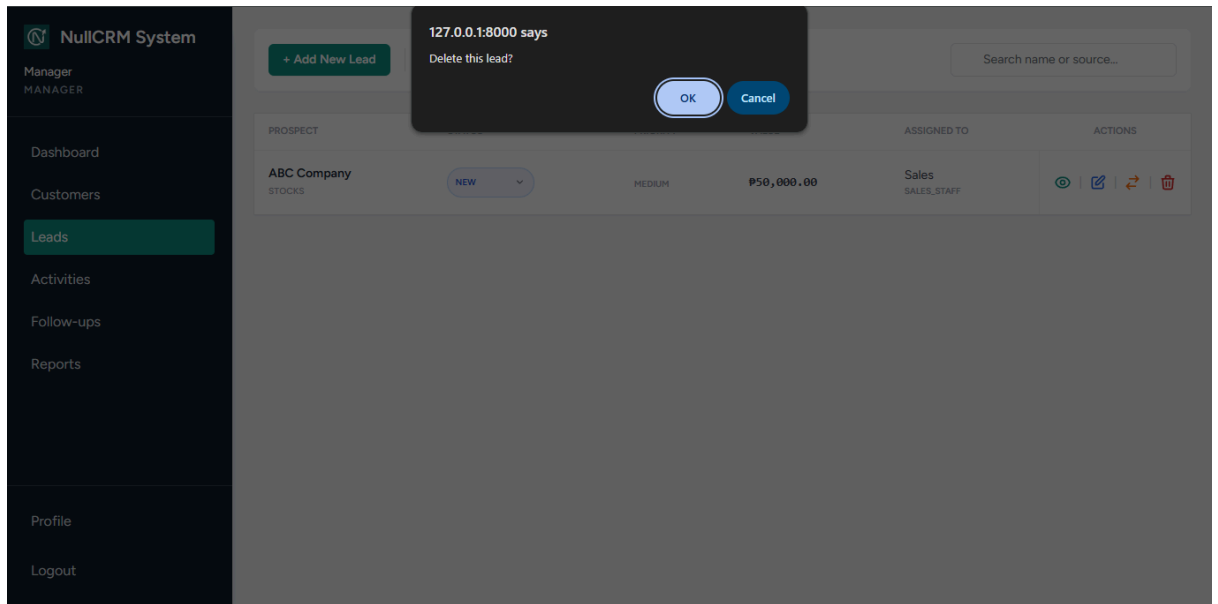
ASSIGNED SALES PERSONNEL

Sales

STAFF USER ID

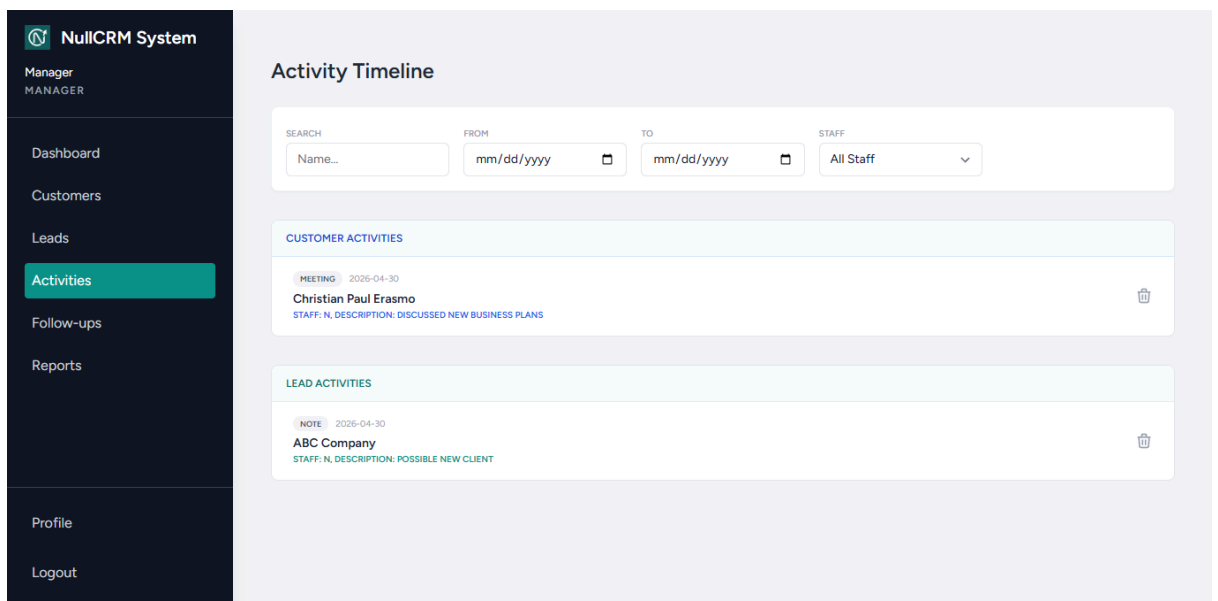
#4

Delete Lead



e. Activity Module

Activity Index



Add Activity

NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

LOG NEW ACTIVITY — LINKED TO CUSTOMER #1

ACTIVITY TYPE *

CALL

DATE *

05/07/2026

DESCRIPTION *

What was discussed?

Cancel

LOG ACTIVITY

Delete Activity

NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

Activity Timeline

127.0.0.1:8000 says
Delete this activity?
OK Cancel

SEARCH

FROM

TO

STAFF

Name...

mm/dd/yyyy

mm/dd/yyyy

All Staff

CUSTOMER ACTIVITIES

MEETING 2026-04-30
Christian Paul Erasmo
STAFF: N, DESCRIPTION: DISCUSSED NEW BUSINESS PLANS

LEAD ACTIVITIES

NOTE 2026-04-30
ABC Company
STAFF: N, DESCRIPTION: POSSIBLE NEW CLIENT

f. Follow-up Module

Follow-up Index

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

+ Add Follow-up

All Staff

LEAD

DUE: 2026-05-02 (OVERDUE)

ABC Company

Have Contact

Try to make contact with the company for business

✓

✎

🗑

STAFF: SALES

ASSIGNED BY: SYSTEM

PENDING

CUSTOMER

DUE: 2026-05-01

Christian Paul Erasmo

Follow up on the new business plan

N/A

🗑

STAFF: MANAGER

ASSIGNED BY: SYSTEM

✓ COMPLETED

Add Follow-up

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

SCHEDULE NEW FOLLOW-UP

SUBJECT / TITLE *

ASSIGN TO LEAD

NONE

ASSIGN TO CUSTOMER

NONE

ASSIGNED STAFF

Please select a Customer or Lead

DUE DATE *

05/07/2026

NOTES (OPTIONAL)

Leave blank for N/A

CANCEL

SAVE FOLLOW-UP

Edit Follow-up

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

EDIT FOLLOW-UP

SUBJECT *
Have Contact

ASSIGN TO LEAD
ABC COMPANY

ASSIGN TO CUSTOMER
NONE

ASSIGNED STAFF
Sales

DUE DATE *
05/07/2026

NOTES
Try to make contact with the company for business

CANCEL

UPDATE FOLLOW-UP

Delete Follow-up

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

+ Add Follow-up

127.0.0.1:8000 says
Delete this follow-up?

OK

Cancel

mm/dd/yyyy

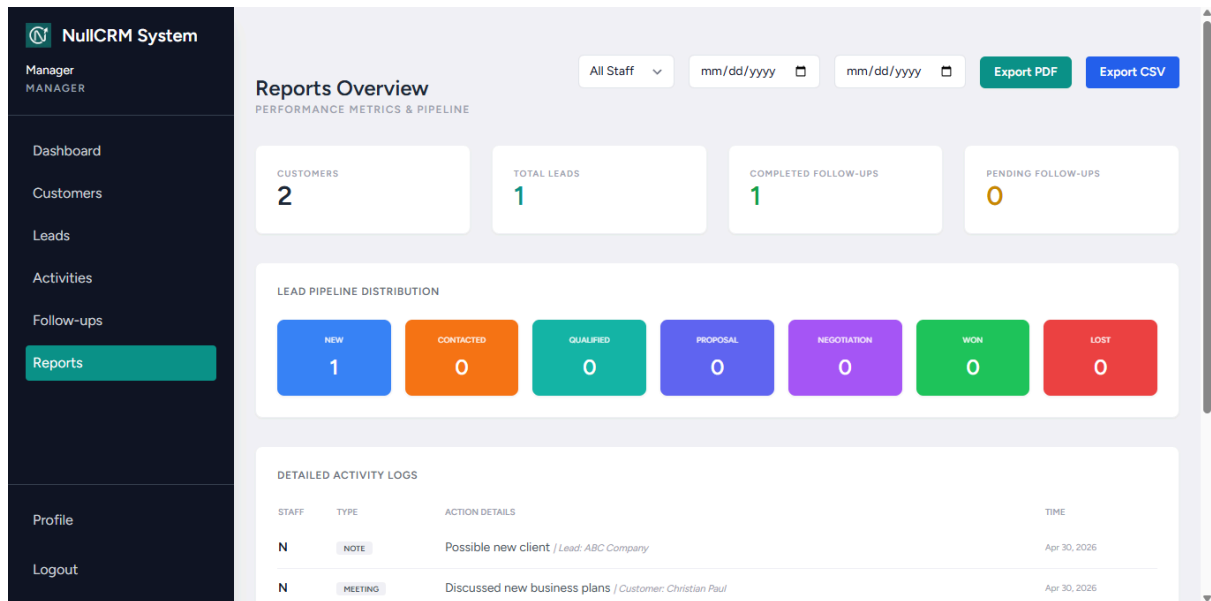
mm/dd/yyyy

All Staff

LEAD DUE: 2026-05-02 (OVER DUE)
ABC Company | Have Contact
Try to make contact with the company for business
STAFF: SALES | ASSIGNED BY: SYSTEM | PENDING

CUSTOMER DUE: 2026-05-01
Christian Paul Erasmo | Follow up on the new business plan
N/A
STAFF: MANAGER | ASSIGNED BY: SYSTEM | COMPLETED

g. Report Module



XI. Testing and Results

The NullCRM system underwent several phases of testing to ensure data integrity, security, and a seamless user experience:

- **CRUD Functional Testing:** Performed rigorous testing on the Customer, Lead, Activity, and Follow-up modules to ensure records can be added, edited, and searched according to the defined validation rules (e.g., required emails and unique phone numbers).

Testing (CRUD on Customer):

Create Customer:

 NullCRM System

N
ADMIN

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

System Archive

Users

Logout

← Add New Customer

CONTACT INFORMATION

FIRST NAME
e.g. John

LAST NAME
e.g. Doe

EMAIL ADDRESS
john.doe@example.com

CONTACT NUMBER
09123456789

COMPANY & CLASSIFICATION

COMPANY NAME
e.g. ABC Corp

STATUS
Active

OFFICE ADDRESS
Street, City, Province

ASSIGN TO
-- Select Assignee --

 NullCRM System

N
ADMIN

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

System Archive

Users

Logout

← Add New Customer

CONTACT INFORMATION

FIRST NAME
Ulysses Kaiser

LAST NAME
Alzate

EMAIL ADDRESS
uka@gmail.com

CONTACT NUMBER
09192504321

COMPANY & CLASSIFICATION

COMPANY NAME
CPE Company

STATUS
Active

OFFICE ADDRESS
Narnia

ASSIGN TO
Sales (Sales staff)

Read Customer:


NullCRM System

N
ADMIN

Dashboard

Customers

 Leads
 Activities
 Follow-ups
 Reports
 System Archive

Users
 Logout

+ Add Customer


 Search customers...

Active: 2

Inactive: 0

▼ ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmio	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	  
Ulysses Kaiser Alzate	uka@gmail.com	09192504321	CPE COMPANY	Sales SALES_STAFF	  

▼ INACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
No inactive contacts found.					

NullCRM System

N
ADMIN

- Dashboard
- Customers**
- Leads
- Activities
- Follow-ups
- Reports
- System Archive
- Users
- Logout

+ Add Customer
Active: 1 Inactive: 0

▼ ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Ulysses Kaiser Alzate	uka@gmail.com	09192504321	CPE COMPANY	Sales SALES_STAFF	

▼ INACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
No inactive contacts found.					

Update Customer:

 NullCRM System

N
ADMIN

[Dashboard](#)

[Customers](#)

[Leads](#)

[Activities](#)

[Follow-ups](#)

[Reports](#)

[System Archive](#)

[Users](#)

[Logout](#)

← Ulysses Kaiser Alzate
CUSTOMER PROFILE • ID #3

ACTIVE

GENERAL INFORMATION

EMAIL ADDRESS
uka@gmail.com

CONTACT NUMBER
09192504321

COMPANY / ORGANIZATION
CPE Company

OFFICE ADDRESS
Narnia

INTERNAL ASSIGNMENT DETAILS

ASSIGNED STAFF NAME
Sales

STAFF USER ID
#4

RECENT ACTIVITIES

+ LOG NEW ACTIVITY

No activities recorded.

RECENT FOLLOW-UPS

+ SCHEDULE NEW

No follow-ups found.

 NullCRM System

N
ADMIN

[Dashboard](#)

[Customers](#)

[Leads](#)

[Activities](#)

[Follow-ups](#)

[Reports](#)

[System Archive](#)

[Users](#)

[Logout](#)

← Edit Customer
Ulysses Kaiser Alzate • ID #3

CONTACT INFORMATION

FIRST NAME
Ulysses Kaiser

LAST NAME
Alzate

EMAIL ADDRESS
uka@gmail.com

CONTACT NUMBER

COMPANY & CLASSIFICATION

COMPANY NAME
CPE Company

ACCOUNT STATUS
Active

OFFICE ADDRESS
Narnia

INTERNAL ASSIGNMENT DETAILS

ASSIGN TO

STAFF ID

NullCRM System

N

ADMIN

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

System Archive

Users

Logout

←

Edit Customer

Ulysses Kaiser Alzate • ID #3

CONTACT INFORMATION

FIRST NAME

Ulysses Kaiser

LAST NAME

Alzate

EMAIL ADDRESS

uka@gmail.com

CONTACT NUMBER

09098152500

COMPANY & CLASSIFICATION

COMPANY NAME

CPE Company

ACCOUNT STATUS

Active

OFFICE ADDRESS

Narnia

INTERNAL ASSIGNMENT DETAILS

ASSIGN TO

STAFF ID

NullCRM System

N

ADMIN

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

System Archive

Users

Logout

+ Add Customer

🔍 Search customers...

Active: 2

Inactive: 0

▼ ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmo	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	
Ulysses Kaiser Alzate	uka@gmail.com	09098152500	CPE COMPANY	Sales SALES_STAFF	

▼ INACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
No inactive contacts found.					

Delete Customer:

The screenshot shows the NullCRM System interface. A dark sidebar on the left contains navigation links: Dashboard, Customers (highlighted), Leads, Activities, Follow-ups, Reports, System Archive, Users, and Logout. The main content area has a top bar with a '+ Add Customer' button, a search bar, and status indicators for 'Active: 2' and 'Inactive: 0'. A modal dialog is open in the center, displaying the IP address '127.0.0.1:8000 says' and the question 'Delete this contact?'. The dialog has 'OK' and 'Cancel' buttons. Below the dialog, there are two sections: 'ACTIVE CONTACTS' and 'INACTIVE CONTACTS'. The 'ACTIVE CONTACTS' section contains a table with two rows of customer data.

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmo	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	
Ulysses Kaiser Alzate	uka@gmail.com	09098152500	CPE COMPANY	Sales SALES_STAFF	

The 'INACTIVE CONTACTS' section is currently empty, displaying the message 'No inactive contacts found.'

This screenshot shows the same NullCRM System interface after a customer has been deleted. The sidebar remains the same. The top bar now shows 'Active: 1' and 'Inactive: 0'. The 'ACTIVE CONTACTS' table now only contains one row for Christian Paul Erasmo. The 'INACTIVE CONTACTS' section remains empty with the message 'No inactive contacts found.'

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmo	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	

- **Soft Delete Verification:** Specific tests were conducted to ensure that deleting a customer or lead did not permanently purge data but instead moved it to the System Archive, while also verifying that related records (activities and follow-ups) remained linked or were appropriately "hidden" from active views.

Testing (Deleting a Customer on Admin):

Before Deleting Ulysses

 NullCRM System

Admin
ADMIN

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

System Archive

Users

Logout

+ Add Customer

 Search customers...

Active: 2

Inactive: 0

▼ ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmo	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	  
Ulysses Kaiser Alzate	uka@gmail.com	09098152500	CPE COMPANY	Sales SALES_STAFF	  

▼ INACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
No inactive contacts found.					

 NullCRM System

Admin
ADMIN

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

System Archive

Users

Logout

Activity Timeline

SEARCH

FROM

TO

STAFF







CUSTOMER ACTIVITIES

CALL 2026-05-08

Ulysses Kaiser Alzate

STAFF: ADMIN, DESCRIPTION: TEST



MEETING 2026-04-30

Christian Paul Erasmo

STAFF: N, DESCRIPTION: DISCUSSED NEW BUSINESS PLANS



LEAD ACTIVITIES

NOTE 2026-04-30

ABC Company

STAFF: N, DESCRIPTION: POSSIBLE NEW CLIENT



127.0.0.1:8000/dashboard

After Deleting Ulysses

 NullCRM System

Admin
ADMIN

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

System Archive

Users

Logout

+ Add Customer

 Search customers...

Active: 1

Inactive: 0

▼ ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmo	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	  

▼ INACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
No inactive contacts found.					

 NullCRM System

Admin
ADMIN

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

System Archive

Users

Logout

Activity Timeline

SEARCH

FROM

TO

STAFF

Name...

mm/dd/yyyy

mm/dd/yyyy

All Staff

CUSTOMER ACTIVITIES

MEETING

2026-04-30

Christian Paul Erasmo

STAFF: N, DESCRIPTION: DISCUSSED NEW BUSINESS PLANS



LEAD ACTIVITIES

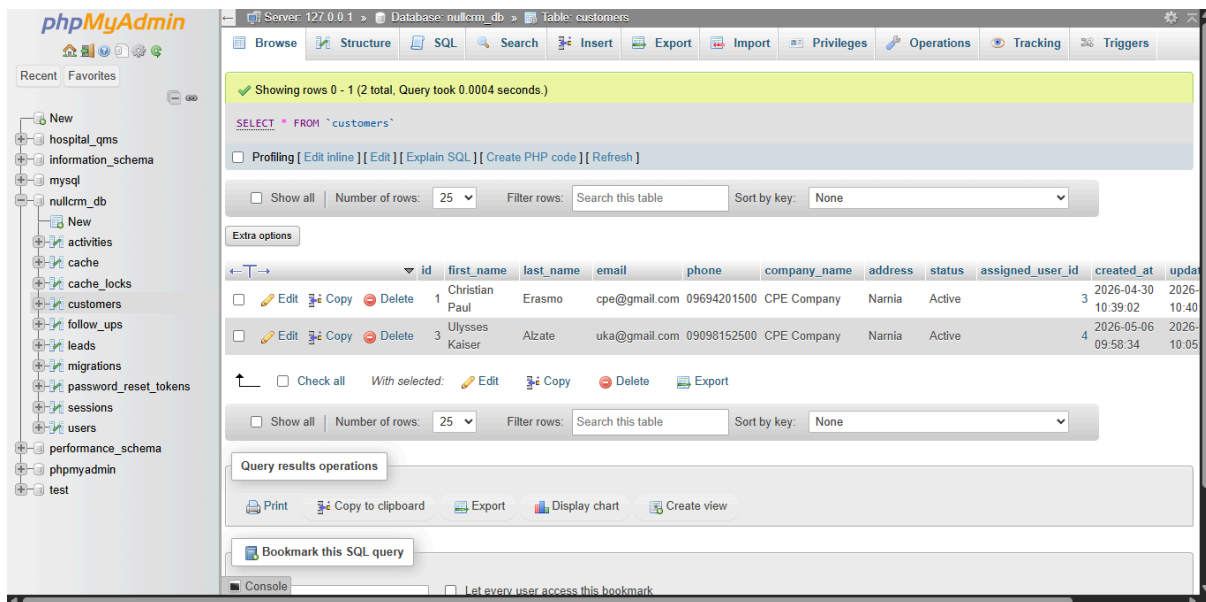
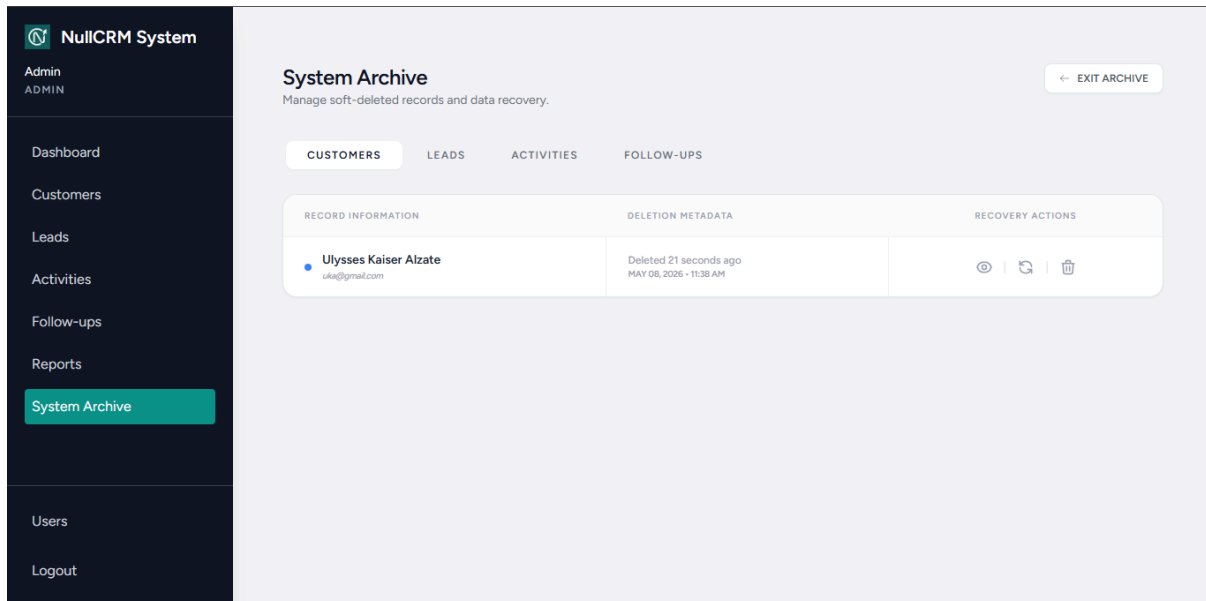
NOTE

2026-04-30

ABC Company

STAFF: N, DESCRIPTION: POSSIBLE NEW CLIENT





- **Role-Based Access Control (RBAC):** We verified that different user roles (Admin, Manager, Staff) saw filtered data relevant only to their permissions, such as Staff only seeing their assigned customers.

Testing (Manager & Sales Staff Customer View):

Manager:

 NullCRM System

Manager
MANAGER

Dashboard

Customers

Leads

Activities

Follow-ups

Reports

Profile

Logout

+ Add Customer

Search customers...

Active: 2

Inactive: 0

▼ ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
Christian Paul Erasmo	cpe@gmail.com	09694201500	CPE COMPANY	Manager MANAGER	  
Ulysses Kaiser Alzate	uka@gmail.com	09098152500	CPE COMPANY	Sales SALES_STAFF	  

▼ INACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ASSIGNED USER	ACTIONS
No inactive contacts found.					

Sales Staff:

 NullCRM System

Sales
SALES_STAFF

Dashboard

Customers

Leads

Activities

Follow-ups

Profile

Logout

+ Add Customer

Search customers...

Active: 1

Inactive: 0

▼ ACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ACTIONS
Ulysses Kaiser Alzate	uka@gmail.com	09098152500	CPE COMPANY	  

▼ INACTIVE CONTACTS

NAME	EMAIL	CONTACT NO	COMPANY	ACTIONS
No inactive contacts found.				

XII. Conclusions and Recommendations

The NullCRM project successfully implemented a centralized web-based application using the Laravel MVC framework. By adhering to the proposed database schema and entity relationships, the system provides a robust platform for managing customer information, sales leads, and interaction history. The integration of Eloquent ORM and Blade templating ensured that the application remains scalable and responsive, meeting both the general objective of organizing data and the specific objective of assigning role-based permissions.

Suggestions for future improvement:

- **External Integration:** While currently out of scope, future versions could benefit from SMS gateway integration to provide automated text reminders for follow-ups.
- **Advanced Data Processing:** Implementation of Advanced AI analytics to predict lead conversion probabilities based on historical activity data.
- **Financial Features:** Expanding the system to include Accounting and Invoicing modules to track the actual deal value beyond the "Expected Value" currently recorded in the Lead module.
- **Enhanced Reporting:** Developing automated report scheduling to email PDF performance summaries to Managers and Admins weekly.